

Zachary Ryan

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EDUCATION

Robert Morris University

Bachelor of Science in Cybersecurity

Outstanding Cybersecurity Student Award [\[Link\]](#) | Dean's List Recipient

Pittsburgh, PA

August 2024- December 2026

GPA: 3.91/4.0

EXPERIENCE

Brixmor Property Group | Cybersecurity Intern | Philadelphia, PA

May 2026 - August 2026

- Developed a risk-based methodology for Cisco Umbrella and assessed 100+ domains, recommending blocking or allowing
- Assisted in creating a Third-Party Risk Management (TPRM) policy for vendor onboarding and risk assessment activities
- Performed vulnerability management on 25+ high-risk assets, documented findings, and recommended remediation measures
- Managed cybersecurity operations across a 500+ user enterprise environment, supporting security investigations and analysis

United States Navy | Machinist Mate - Nuclear Reactor Department | Norfolk, VA

July 2020 - September 2023

- Functioned under high-pressure conditions while maintaining strict attention to detail and compliance with safety standards
- Handled incident response, system monitoring, diagnostics, and secondary nuclear reactor equipment troubleshooting
- Maintained secure reactor operations by enforcing layered access controls, reviewing system status logs, and detecting abnormal system behavior, ensuring operational continuity, technical resilience, and safety in mission-critical environments

PUBLICATIONS

Ryan, Z., & Mishra, S. (2026). *Auditable Evidence in AI Cybersecurity Compliance: A Cross-Framework Analysis*.

Computers & Security (Submitted June 2026)

- Conducted a cross-framework analysis of NIST AI RMF, ISO/IEC 42001, EU AI Act, CSA AICM, AWS, and Google Cloud
- Analyzed how governance, regulatory, control, and vendor-specific sources define, imply, or omit auditable evidence
- Evaluated compliance evidence across documentation, logging, monitoring, access control, model security, and testing
- Identified gaps between AI cybersecurity requirements, audit evidence, and assurance sufficiency across major frameworks

Ryan, Z., & Mishra, S. (2026). *Evaluating AI vs Human Phishing Evasion Across Emulated Email Filters and SpamAssassin*.

NEDSI Annual Conference (Accepted; presented April 2026) [\[Publication Link - PDF pp. 189 - 202\]](#)

- Created controlled prompts to generate realistic AI phishing emails using dummy URLs and fictional identities
- Built a 1,260-email dataset comparing AI and human phishing across account, financial, and delivery themes
- Applied Chi-square and Z-tests showing AI phishing evaded selected emulated filters more often than human phishing emails
- Developed a Python-based filter emulation framework to score phishing emails, compare AI versus human bypass rates, evaluate provider-specific thresholds, and test emulated Gmail, Outlook, Yahoo, and SpamAssassin detection conditions

PROJECTS

Policy-as-Code Docker Network Segmentation Lab | [\[Github\]](#)

- Designed a segmented enterprise network architecture with public, private, and management security zones
- Implemented policy-as-code controls and firewall enforcement mechanisms to restrict unauthorized network access
- Developed automated testing and visualization tools to validate security policy compliance across network segments

AI Powered Phishing Email Detection & Safe Rewrite System | [\[Github\]](#)

- Built a full-stack phishing detection platform using FastAPI and React to classify emails as SAFE, SUSPICIOUS, or PHISHING
- Implemented explainable risk scoring logic using keyword analysis, links, and SPF/DKIM/DMARC email authentication headers
- Developed a safe email rewrite engine that removes malicious content and highlights changes using HTML diff visualization

SOC-Style Network Anomaly Detection Lab | [\[Github\]](#)

- Built a full-stack SOC-style network anomaly detection platform using FastAPI, Docker, and React to analyze traffic patterns
- Implemented unsupervised machine learning (Isolation Forest) to score and flag anomalous real-time network traffic
- Developed a traffic simulation engine generating traffic flows, attacker behavior, and port scan anomalies

SKILLS

Languages: Python | Java | Bash | PowerShell | HTML | CSS | SQL

Technical: Linux | Docker | Git/GitHub | TCP/IP | Firewalls | Operating Systems | Digital Forensics | Network Segmentation

Security Tools: CrowdStrike Falcon | Rapid7 | Azure | NinjaOne | Mimecast | Okta | Keeper | FreshService

Governance & Policy: NIST AI RMF | ISO/IEC 42001 | AI Governance | Cyber Policy | Auditability | Risk Management

Core Competencies: Research | Technical Writing | Critical Thinking | Leadership | Attention to Detail

LICENSES AND CERTIFICATIONS

- DoD Secret Clearance | CompTIA Security+ (Scheduled) | Google Cybersecurity Professional Certificate [\[Link\]](#)